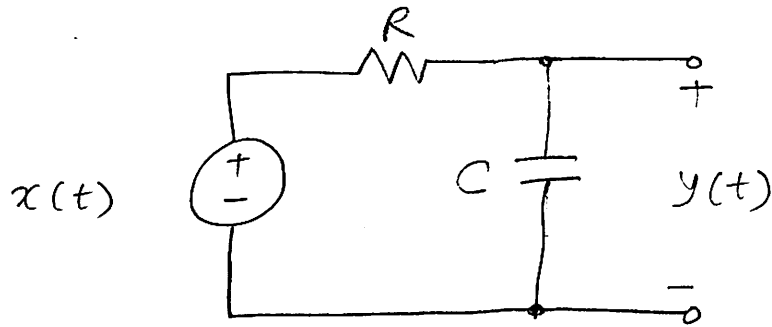


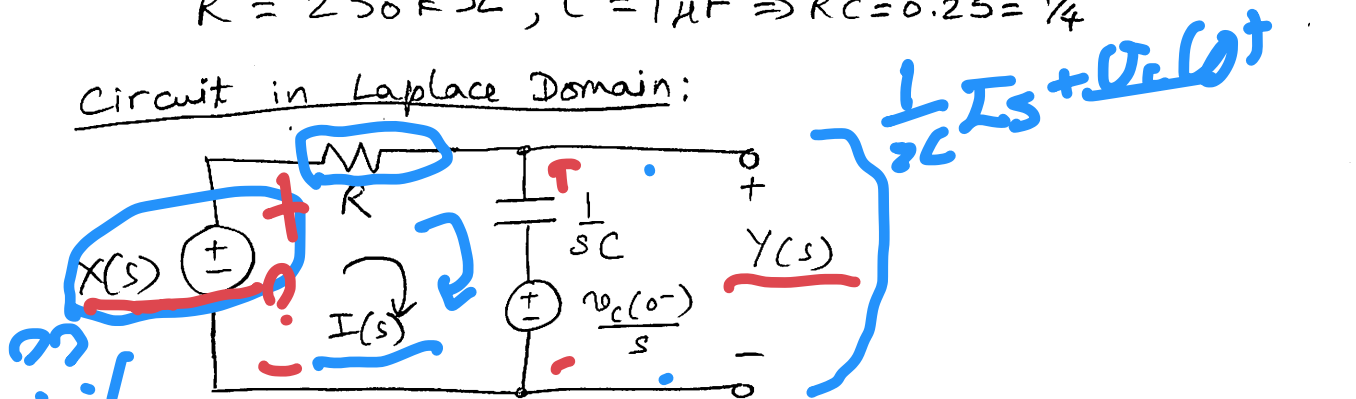


$$x(t) = 25 \cos(3t) \Rightarrow X(s) = 25 \times \frac{s}{s^2 + 3^2} = \frac{25s}{(s^2 + 9)}$$

$$v_c(0^-) = 2V$$

$$R = 250 \text{ k}\Omega, C = 1 \mu\text{F} \Rightarrow RC = 0.25 = \frac{1}{4}$$

Circuit in Laplace Domain:



$$Y(s) = \frac{v_c(0^-)}{s} + \frac{1}{sC} I(s)$$

$$I(s) = \frac{X(s) - \frac{v_c(0^-)}{s}}{R + \frac{1}{sC}} = \frac{sC X(s) - v_c(0^-) \cancel{sC}}{(sCR + 1) \cancel{sC}}$$

$$= \frac{sC}{(sCR + 1)} X(s) - v_c(0^-) \times \frac{C}{(1 + sCR)}$$

$$\Rightarrow Y(s) = \frac{v_c(0^-)}{s} + \frac{1}{sC} \left\{ \frac{sC X(s)}{(sCR + 1)} - v_c(0^-) \times \frac{C}{(1 + sCR)} \right\}$$

$$= \frac{X(s)}{(1 + sCR)} + v_c(0^-) \left\{ \frac{1}{s} - \frac{1}{s(1 + sCR)} \right\}$$

$$Y(s) = \frac{X(s)}{(1+sCR)} + v_c(0^-) \times \frac{1}{s(1+sCR)}$$

$$= \underbrace{\frac{X(s)}{(1+sCR)}}_{\text{zero state response}} + \underbrace{v_c(0^-) \times \frac{CR}{(1+sCR)}}_{\text{zero input response}} = \frac{4X(s)}{(s+4)} + v_c(0^-) \times \frac{1}{(s+4)} //$$

(a) zero input response:

$$Y_{ZIR}(s) = \frac{v_c(0^-)RC}{1+sRC} = \frac{0.25 \times 2}{1+s \times 0.25} = \frac{1 \times 2}{(s+4)}$$

$$= 2e^{-4t} u(t) //$$

(b) zero state response:

$$Y_{ZSR}(s) = \frac{X(s)}{1+sCR} = \frac{4X(s)}{(4+s)} = \frac{4 \times 25s}{(s+4)(s^2+9)}$$

$$= \frac{100s}{(s+4)(s^2+9)}$$

Need to do a partial fraction expansion to invert this:

$$\frac{100s}{(s+4)(s^2+9)} = \frac{A_1}{s+4} + \frac{A_2}{s+j3} + \frac{A_3}{s-j3}$$

$$A_1 = \left. \frac{100s}{s^2+9} \right|_{s=-4} = \frac{-400}{(16+9)} = \frac{-400}{25} = -16 //$$

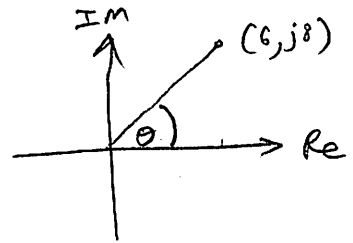
$$A_2 = \frac{100s}{(s+4)(s-j3)} \Big|_{s=-j3} = \frac{-300j}{(4-j3)(-j3-j3)} = \frac{(-300j)}{(4-j3)(-j6)}$$

$$= \frac{50}{4-j3} = \frac{50 \times (4+j3)}{(4-j3) \times (4+j3)} = \frac{200 + j150}{(16+9)} = 8 + j6 //$$

$$A_3 = A_2^* = 8 - j6 //$$

$$|A_2| = \sqrt{8^2 + 6^2} = \sqrt{100} = 10$$

$$\angle A_2 = \theta = \tan^{-1}\left(\frac{6}{8}\right) = 36.9^{\circ} //$$



$$Y_{ZSR}(s) = \frac{-16}{s+4} + \frac{10e^{+j\theta}}{s+j3} + \frac{10e^{-j\theta}}{s-j3}$$

$$\begin{aligned} \Rightarrow y_{ZSR}(t) &= [-16e^{-4t} + 10e^{+j\theta} \times e^{-j3t} + 10e^{-j\theta} \times e^{j3t}] u(t) \\ &= [-16e^{-4t} + 20 \left(\frac{e^{-j(3t-\theta)}}{2} + \frac{e^{+j(3t-\theta)}}{2} \right)] u(t) \\ &= [-16e^{-4t} + 20 \cos(3t-\theta)] u(t) // \end{aligned}$$

(C) Transient Response

Now,

$$Y(s) = 4 \frac{X(s)}{(s+4)} + u_c(0^-) \times \frac{1}{(s+4)} = \frac{4 \times 25s}{(s+4)(s^2+9)} + \frac{2}{s+4}$$

$$= \frac{-16}{s+4} + \frac{10e^{j\theta}}{s+j3} + \frac{10e^{-j\theta}}{s-j3} + \frac{2}{s+4} = \frac{-14}{s+4} + \frac{10e^{j\theta}}{s+j3} + \frac{10e^{-j\theta}}{s-j3}$$

$$\Rightarrow y(t) = \underbrace{[-14e^{-4t}]_{\text{Transient}}} + \underbrace{20 \cos(3t-\theta)}_{\text{steady state}} u(t)$$

$$\Rightarrow y_{\text{Transient}}(t) = -14e^{-4t}u(t) //$$

(d) Steady-state Response:

$$\Rightarrow y_{\text{Steady state}}(t) = 20 \cos(3t - \theta)u(t) //$$

where $\theta = 36.9^\circ$

We can also partition $y(t)$ into a $y_{\text{natural}}(t) + y_{\text{forced}}(t)$. The natural response is like basically the impulse response $h(t)$.

$$y_{\text{natural}}(t) = -14e^{-4t}u(t) //$$

$$y_{\text{forced}}(t) = 20 \cos(3t - \theta)u(t) //$$